

A Conformally Symplectic Damped Hénon Dimer: Exact Period-Two Mode Monodromy

Route-A finite certificate C118

Abstract

We construct an exact rational certificate for a damped two-site variational Hénon map. Its Jacobian is conformally symplectic with factor $1/2$ and determinant $1/4$. Two synchronous fixed points and one primitive synchronous period-two orbit are verified exactly. The two-step monodromy factors into longitudinal and transverse Laplacian modes with traces $-59/4$ and -13 . An uncoupled control isolates a trace shift of $7/4$. The result is a finite low-period tangent witness, not a global transfer-operator theorem.

1 Damped variational model

For $q, p \in \mathbb{Q}^2$, let

$$U(q) = \sum_{i=1}^2 \left(\frac{13}{4} q_i^2 - \frac{1}{3} q_i^3 \right) - \frac{1}{8} (q_1 - q_2)^2, \quad F(q, p) = (\nabla U(q) - \tfrac{1}{2} p, q).$$

Writing

$$L = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad H(q) = \text{diag}(13/2 - 2q_i) - \tfrac{1}{4} L,$$

the Jacobian and its exact inverse are

$$J(q) = \begin{pmatrix} H(q) & -I_2/2 \\ I_2 & 0 \end{pmatrix}, \quad F^{-1}(Q, P) = (P, 2(\nabla U(P) - Q)).$$

For $\Omega = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$, direct multiplication gives

$$J(q)^\top \Omega J(q) = \tfrac{1}{2} \Omega, \quad \det J(q) = \tfrac{1}{4}.$$

There is also an exact primitive relation. With $\lambda = q \cdot dp$,

$$F^* \lambda - \tfrac{1}{2} \lambda = d(U(q) - \tfrac{1}{2} p \cdot q).$$

These are polynomial identities, not numerical sample fits.

2 Exact synchronous orbit witnesses

Synchrony is invariant because $L(1, 1)^\top = 0$. The fixed states are

$$P_0 = ((0, 0), (0, 0)), \quad P_5 = ((5, 5), (5, 5)).$$

The two distinct states

$$C_2 = ((2, 2), (6, 6)), \quad C_6 = ((6, 6), (2, 2))$$

satisfy $F(C_2) = C_6$ and $F(C_6) = C_2$, giving a primitive period-two witness. The direct four-dimensional two-step monodromy is

$$M = \begin{pmatrix} -107/8 & -7/8 & 23/8 & -1/8 \\ -7/8 & -107/8 & -1/8 & 23/8 \\ 9/4 & 1/4 & -1/2 & 0 \\ 1/4 & 9/4 & 0 & -1/2 \end{pmatrix},$$

with trace $-111/4$ and determinant $1/16$.

3 Longitudinal and transverse factors

The Laplacian eigenvalues are 0 and 2. At synchronous coordinate s , the scalar Hessian value in mode ℓ is

$$h_s(\ell) = 13/2 - 2s - \ell/4.$$

Thus the mode pairs at $s = 2, 6$ are $(5/2, -11/2)$ and $(2, -6)$. A one-step mode matrix is $J_h = \begin{pmatrix} h & -1/2 \\ 1 & 0 \end{pmatrix}$. The exact two-step data are

mode	(h_2, h_6)	$\text{tr}(J_{h_6} J_{h_2})$	determinant
longitudinal	$(5/2, -11/2)$	$-59/4$	$1/4$
transverse	$(2, -6)$	-13	$1/4$

Consequently the direct and mode reconstructions agree:

$$\begin{aligned} \det(I - zM) &= (1 + 59z/4 + z^2/4)(1 + 13z + z^2/4) \\ &= 1 + \frac{111}{4}z + \frac{769}{4}z^2 + \frac{111}{16}z^3 + \frac{1}{16}z^4. \end{aligned}$$

At $\kappa = 0$, the synchronous orbit remains valid but both modes are longitudinal: the full trace is $-59/2$. Coupling therefore shifts the direct trace by $7/4$ in this controlled tangent comparison.

4 Reproducibility and scope

From the package directory run

```
python3 code/c118_damped_dimer_producer.py
python3 code/c118_damped_dimer_checker.py
python3 code/c118_sympy_crosscheck.py
python3 code/c118_replay.py
python3 code/c118_mutation.py
```

The independent checker, 13 symbolic checks, canonical replay, and all 12 hostile mutations pass.

The verdict is $A1 = \mathbf{A1_WEAK}$: only named fixed and period-two witnesses are certified. $A2 = \mathbf{A2_FAIL}$ because tangent monodromy is not a source-owned transfer operator; $A3$ is not addressed and $A4$ fails. We claim no complete orbit atlas, transfer/Fredholm/nuclear owner, arithmetic local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Route B. The firewall is $\mathbf{NO_BAD_EULER_OR_ROOT_NUMBER}$.